
MB502T

(THEORY)

Collected Exam Questions

With Comprehensive Answers



Jan 29 – Feb 18, 2026

"When we lift others, Allah lifts us." 🤝✌️

Prepared by: **M. Kashif**

Program: BS Zoology

University: University of Okara, Pakistan



03064963933



burhan.wani888@gmail.com



WhatsApp Group: <https://chat.whatsapp.com/LOsovr0yLIm6OtYldfW1jT>

Total Questions: 106 | Complete Answers Included



QUESTIONS & ANSWERS



Q1. What is elongation polymerase? / Polymerase in elongation phase of transcription?

Answer:

The **elongation polymerase** is the RNA polymerase that makes the RNA chain during the elongation step of transcription.

In bacteria, the sigma factor leaves after starting transcription, and the remaining core enzyme carries on making the RNA.

In eukaryotes, RNA Polymerase II does this job. Its **CTD tail** gets a phosphate group added at Serine-2, which helps it slide along the DNA and bring in helper proteins.

The enzyme reads the DNA template from **3' to 5'** and builds the new RNA strand from **5' to 3'**, adding one nucleotide at a time.

Q2. Name any two features of replacement vectors.

Answer:

1. They have a middle piece of DNA called a **stuffer fragment** that is not needed. This piece is cut out and replaced with the DNA we want to clone. This allows us to insert large pieces of DNA (up to 23 kb).
2. They have **two cutting sites** for the same restriction enzyme on either side of the stuffer fragment, so it can be easily removed and swapped with foreign DNA.

Q3. What is Class II release factor (RF3) and its function?

Answer:

RF3 is a Class II release factor. It is a protein that binds to GTP and helps during the last step of translation (making proteins).

Function: RF3 does not read stop codons itself. Its job is to **remove the Class I release factors** (RF1 or RF2) from the ribosome after the protein chain has been released. It uses the energy from GTP to change the shape of the ribosome and push RF1/RF2 off, so the ribosome can be recycled.

Q4. What is Class I release factor (RF1/RF2) and its function?

Answer:

Class I release factors are proteins that read stop codons on the mRNA during protein-making.

RF1 reads the stop codons **UAA and UAG**. **RF2** reads the stop codons **UAA and UGA**.

Function: When a stop codon comes into the A-site of the ribosome, RF1 or RF2 sits there instead of a tRNA. They cause the bond between the protein chain and the tRNA to break, so the finished protein is released from the ribosome.

Q5. During RNA editing, what effect happens on deamination code? (C to U or A to I)**Answer:**

RNA editing by deamination changes the message in the mRNA after it has been copied from DNA.

C to U editing: An enzyme called **cytidine deaminase** removes an amino group from cytosine and turns it into uracil. For example, in the apolipoprotein B gene, a C is changed to U in the intestine, which creates a new stop codon. This makes a shorter protein (apoB-48) instead of the full protein (apoB-100) made in the liver.

A to I editing: An enzyme called **ADAR** changes adenosine to inosine by removing an amino group. The ribosome reads inosine as guanosine (G), so the protein made may have a different amino acid in it.

Q6. Gene expression by regulatory proteins – A-site, P-site, E-site.**Answer:**

Gene expression is regulated at the translation level through the three sites of the ribosome:

A-site (Aminoacyl site): The new charged tRNA (carrying an amino acid) enters here. The ribosome checks if the tRNA matches the mRNA codon at this site. Regulatory proteins can block this site to slow down or stop protein making.

P-site (Peptidyl site): This holds the tRNA that is carrying the growing protein chain. The new amino acid from the A-site is joined to this chain here.

E-site (Exit site): After the tRNA has given away its amino acid, it moves here and then leaves the ribosome.

Regulatory proteins can speed up or slow down how the ribosome works at these sites, and this is one way cells control gene expression.

Q7. What are A, P, E sites in ribosomes?**Answer:**

The ribosome has three special sites where tRNA molecules sit during protein making:

A-site (Aminoacyl site): This is on the right side. The new tRNA with its amino acid comes in here. The ribosome checks if the tRNA anticodon matches the mRNA codon.

P-site (Peptidyl site): This is in the middle. It holds the tRNA that has the growing protein chain attached to it.

E-site (Exit site): This is on the left side. It holds the empty tRNA (which has already given its amino acid) just before it leaves the ribosome.

The tRNAs move through these sites in order: **A → P → E**.

Q8. Functions of A-site, P-site, E-site?**Answer:**

A-site: Receives the new tRNA that is carrying an amino acid. The matching of the codon and anticodon happens here. This is where the correct amino acid is picked.

P-site: Holds the tRNA with the growing chain. The **peptide bond** (link between amino acids) is made here when the chain moves from the P-site tRNA to the amino acid on the A-site tRNA.

E-site: Holds the empty tRNA that has already given its amino acid. The tRNA leaves the ribosome from this site so it can be recharged with a new amino acid.

Q9. Regulatory proteins functions in gene expression.

Answer:

Regulatory proteins control which genes are turned on or off, and how much protein is made:

Activators: These sit on special DNA regions called enhancers and help RNA polymerase to start copying the gene faster.

Repressors: These sit on the operator or silencer regions and block RNA polymerase, so the gene stays switched off.

Transcription factors: These are needed to set up the copying machinery at the promoter so transcription can begin.

Coactivators and corepressors: These do not touch DNA directly. They work with other proteins to fine-tune how much of a gene is expressed.

Q10. How is gene expression controlled by regulatory proteins?

Answer:

Regulatory proteins control gene expression at many levels:

At transcription: Activators help RNA polymerase bind and start copying the gene. Repressors block the promoter so the gene cannot be copied. For example, the lac repressor blocks the lac operon when there is no lactose.

After transcription: Some proteins affect how stable the mRNA is, how it is spliced, and how it is carried out of the nucleus.

At translation: Some proteins sit on the mRNA and either help or stop the ribosome from reading it.

After translation: Proteins can be changed (by adding chemical groups), folded, or broken down to control their activity.

Q11. Regulatory proteins expression – explain.

Answer:

Regulatory proteins are made from regulatory genes. The cell controls how and when these proteins are produced.

Constitutive expression: Some regulatory proteins (like repressors) are made all the time at a low, steady level. They become active or inactive when small signal molecules (like inducers) bind to them.

Conditional expression: Other regulatory proteins are only made when the cell needs them, such as during stress or when a specific nutrient appears.

Feedback loops keep the levels balanced. For example, a regulatory protein may switch off its own gene once there is enough of it in the cell.

Q12. Describe RNA processing events (Capping, Splicing, Polyadenylation).

Answer:

In eukaryotes, the raw mRNA (pre-mRNA) must be processed before it can be used. There are three main steps:

1. 5' Capping: A special cap made of 7-methylguanosine (m7G) is added to the front (5' end) of the mRNA. This cap protects the mRNA from being broken down, helps the ribosome recognise it, and helps it move out of the nucleus.

2. Splicing: The non-coding parts (introns) are cut out, and the coding parts (exons) are joined together. This is done by a complex called the **spliceosome**, which is made of snRNPs.

3. 3' Polyadenylation: A long tail of adenine bases (about 200 A's) called the **poly-A tail** is added to the back (3' end) of the mRNA. The signal AAUAAA tells the cell where to cut and add this tail. The poly-A tail protects the mRNA and helps it get translated.

Q13. How does beta-galactosidase help in the metabolism of lactose?

Answer:

Beta-galactosidase is an enzyme made by the **lacZ gene** in the lac operon of E. coli bacteria.

Main function: It breaks lactose (a sugar made of two parts) into **glucose and galactose** by cutting the bond between them.

It also changes some lactose into **allolactose**, which is very important because allolactose is the signal molecule that switches on the lac operon. Allolactose binds to the repressor protein and pulls it off the DNA, so the genes can be copied.

Q14. What is an Open Reading Frame (ORF)?

Answer:

An Open Reading Frame (ORF) is a stretch of DNA that starts with a **start codon (ATG)** and ends with a **stop codon (TAA, TAG, or TGA)**, with no stop codons in between.

It is basically a sequence that could code for a protein. It must be long enough (usually at least 100 codons) to be a possible gene.

Importance: Finding ORFs in a genome helps scientists predict where the genes are. Not every ORF is a real gene, but every real gene has an ORF.

Q15. What is Shine-Dalgarno sequence?

Answer:

The Shine-Dalgarno sequence is a short sequence (usually **AGGAGG**) found on bacterial mRNA, about 5–10 bases before the start codon (AUG).

Function: It helps place the ribosome in the right position on the mRNA. It does this by pairing with a matching sequence on the **16S rRNA** of the small ribosomal subunit (30S).

This makes sure translation starts at the correct AUG codon. Eukaryotes do not have this sequence. They use the **5' cap and Kozak sequence** instead.

Q16. What is alternative splicing and its importance?

Answer:

Alternative splicing means that one pre-mRNA can be cut and joined in different ways to make different mRNAs from the same gene.

By including or leaving out different exons, the cell can make many different versions of a protein from just one gene.

Importance: This greatly increases protein variety without needing more genes. For example, one gene in fruit flies (DSCAM) can make over **38,000 different mRNAs**. It also lets different tissues make different versions of the same protein.

Q17. What is the sigma factor?

Answer:

The sigma factor is a protein that joins the bacterial RNA polymerase and helps it start transcription.

It attaches to the core enzyme to form the complete enzyme (**holoenzyme**). The sigma factor's job is to find and bind to the **promoter region** on the DNA (the -10 and -35 regions).

After transcription has started and about 8–10 RNA bases are made, the sigma factor **leaves** and the core enzyme continues making the RNA on its own.

The most common sigma factor in E. coli is **sigma-70**, which helps copy everyday housekeeping genes.

Q18. Role/function of sigma factor in stationary phase of bacterial growth/cells.

Answer:

During the stationary phase, bacteria run out of food and stop growing. They switch to a different sigma factor called **sigma-38 (RpoS)**.

Function: Sigma-38 tells the RNA polymerase to copy **stress survival genes**, such as genes for DNA repair, protection against starvation, and resistance to harsh conditions.

By switching from **sigma-70** (used during normal growth) to **sigma-38**, the bacteria can quickly change which genes are active and survive tough times.

Q19. Describe five points of an operon model.

Answer:

The operon model was proposed by Jacob and Monod. It explains how genes are controlled in bacteria. **The five key points are:**

- 1. Structural genes:** A group of genes that code for related proteins. They are read together as one long mRNA. For example, lacZ, lacY, and lacA in the lac operon.
- 2. Promoter:** The region on DNA where RNA polymerase attaches to start copying the genes.
- 3. Operator:** A short piece of DNA between the promoter and the genes. When the repressor protein sits here, it blocks RNA polymerase.
- 4. Regulatory gene:** A separate gene that makes the repressor protein. It can be near or far from the operon.
- 5. Repressor protein:** A protein made by the regulatory gene. When active, it sits on the operator and stops the genes from being copied. When an inducer (like allolactose) binds to it, it changes shape and falls off, so the genes can be read.

Q20. What is the attachment of amino acid in tRNA? / What is the process of aminoacylation?

Answer:

Aminoacylation (or tRNA charging) is the process of attaching the right amino acid to its matching tRNA. It is done by enzymes called **aminoacyl-tRNA synthetases**.

It happens in two steps:

Step 1 – Activation: The amino acid joins with ATP to form an activated amino acid (aminoacyl-AMP). A molecule called pyrophosphate (PPi) is released.

Step 2 – Transfer: The activated amino acid is moved onto the **3' end (CCA tail)** of the correct tRNA. A bond is formed between the amino acid and the last adenine of the tRNA.

There are **20 different synthetase enzymes**, one for each amino acid. Each one makes sure the right amino acid goes on the right tRNA. This is very important for making the correct protein.

Q21. Explain the elongation phase of transcription.

Answer:

Elongation is the stage where RNA polymerase moves along the DNA and builds the RNA strand.

Key points:

- 1.** The enzyme moves along the template DNA strand from **3' to 5'** and makes the RNA from **5' to 3'**.
- 2.** A small bubble (about 12–14 base pairs) opens up in the DNA. About 8–9 bases of the new RNA stay paired with the template inside this bubble.
- 3.** RNA building blocks (ATP, GTP, CTP, UTP) are added one at a time to the growing RNA chain.
- 4.** In bacteria, the sigma factor has already left. In eukaryotes, the **CTD of RNA Pol II** gets phosphorylated at Serine-2 to help with elongation.
- 5.** The speed is about **40–80 nucleotides per second** in bacteria.

Q22. What is self-splicing? Define alternative splicing. What is meant by splicing pathway?**Answer:**

Self-splicing: This is when an intron removes itself from the RNA without needing any protein help. The intron acts as its own enzyme (**ribozyme**). **Group I introns** use an outside guanosine to help cut. **Group II introns** use an internal adenosine and form a loop called a lariat.

Alternative splicing: This is when the same pre-mRNA is cut and joined in different patterns to make different mRNAs. By keeping or removing different exons, one gene can produce many different proteins.

Splicing pathway: This is the step-by-step process of removing introns and joining exons. In the main pathway, **snRNPs (U1, U2, U4, U5, U6)** come together on the pre-mRNA, form a loop at the branch point, cut at both ends of the intron, remove it, and join the exons together.

Q23. Why does mRNA have a short life span? What experiment is used to analyze it?**Answer:**

mRNA has a short life because the cell needs to quickly change which proteins it is making. If mRNA lasted too long, the cell could not respond fast to new signals.

In bacteria, mRNA lasts only **2–5 minutes**. In eukaryotes, it can last from minutes to hours.

mRNA is broken down by enzymes called **RNases** that attack it from the 5' end (after removing the cap) or from the 3' end (after removing the poly-A tail).

Experiment – Pulse-Chase: Cells are given radioactive uridine for a short time (pulse) which gets built into new RNA. Then the label is taken away (chase). Scientists then measure how fast the labelled mRNA disappears over time. The time it takes for half the mRNA to break down is its **half-life**. Northern blotting or RT-qPCR can also be used.

Q24. Process of mRNA maturation?**Answer:**

mRNA maturation is the set of changes that turn the raw pre-mRNA into a finished mRNA ready for use:

1. 5' Capping: A special cap (7-methylguanosine) is added to the front end of the mRNA while it is still being made. This protects it and helps with translation.

2. Splicing: The non-coding bits (introns) are cut out and the coding bits (exons) are joined together by the **spliceosome**.

3. 3' Polyadenylation: The mRNA is cut at the poly-A signal (AAUAAA) and a tail of about **200 adenine bases** is added to the end.

4. Export: The finished mRNA is carried from the nucleus to the cytoplasm through **nuclear pores** with the help of transport proteins.

Only after all these steps can the ribosome use the mRNA to make a protein.

Q25. Why are plasmids not included in the cell's main genome?

Answer:**Plasmids stay separate from the main genome because:**

1. They have their own starting point for copying (**origin of replication**), so they copy themselves on their own.
2. They do not carry genes that are needed for daily survival. The chromosome already has all the essential genes.
3. They carry extra useful genes (like antibiotic resistance) that only help in special situations.
4. They are much smaller than the chromosome and are usually circular in shape.
5. They can be gained or lost without killing the cell, unlike the main chromosome.

Q26. What are the requirements for high-level transcription?**Answer:****For a gene to be copied at a high rate, the cell needs:**

1. **Strong promoter:** A promoter that closely matches the ideal sequence so RNA polymerase can bind easily.
2. **Transcription factors:** Helper proteins (like TFIID, TFIIB, TFIIF in eukaryotes) that set up the machinery at the promoter.
3. **Activator proteins:** These bind to enhancer regions and pull in more helper proteins to speed up transcription.
4. **Open DNA:** In eukaryotes, the DNA must be loose (not tightly packed). **Histone acetylation** and remodelling complexes open up the chromatin.
5. **Enough materials:** The cell must have enough RNA polymerase and nucleotides (ATP, GTP, CTP, UTP) available.

Q27. How does nucleosome help transcription machinery bind to promoter?**Answer:**

Nucleosomes normally wrap DNA tightly and can block the promoter. But the cell has ways to move them out of the way:

1. **Remodelling complexes** (like SWI/SNF) use energy from ATP to slide or push nucleosomes away from the promoter, leaving the DNA open.
2. **Acetylation:** Enzymes called **HATs** add acetyl groups to histone tails. This loosens the grip of histones on DNA.
3. **Histone variants:** Special histones (like H2A.Z) can replace normal ones and make the nucleosome easier to move.
4. **Activator proteins** call in these remodelling complexes to specific promoters, making sure only the right genes get opened up.

Q28. What is RNA editing?**Answer:**

RNA editing is a process that changes the sequence of an mRNA after it has been copied from DNA. This means the final mRNA is different from the original DNA code.

There are two main types:

1. Base changes – such as C-to-U editing (by a deaminase enzyme) and A-to-I editing (by ADAR enzyme). These change single bases in the mRNA.

2. Insertion/deletion editing – where extra uridine bases are added or removed from the mRNA. This is common in trypanosome parasites and is guided by small guide RNAs.

RNA editing increases the variety of proteins a cell can make without changing its DNA.

Q29. Molecular process of RNA editing?

Answer:

The molecular steps depend on the type of editing:

C-to-U editing: A specific enzyme (**APOBEC**) finds a certain cytosine in the mRNA using a nearby signal sequence. It removes an amino group from C and turns it into U. For example, this happens in the ApoB mRNA in the intestine.

A-to-I editing: The **ADAR enzyme** finds double-stranded parts of the RNA. It removes an amino group from adenosine and turns it into inosine. The ribosome reads inosine as guanosine (G), so the protein may change.

Insertion/deletion editing (in trypanosomes): Small **guide RNAs** pair with the mRNA. Where there are mismatches, the mRNA is cut, uridines are added or removed, and the mRNA is sealed back together by a ligase enzyme.

Q30. Explain mechanism of RNA editing?

Answer:

RNA editing changes the mRNA message after it is made:

C-to-U editing: The enzyme complex finds the target C using a nearby guide sequence. It removes the amino group from cytidine, turning it into uridine. This can create new stop codons or change what amino acid is coded.

A-to-I editing: ADAR enzyme binds to areas where the RNA folds into a double strand. It changes adenosine to inosine. Since inosine acts like guanosine during translation, the protein coded by the mRNA may be different.

Insertion/deletion editing: In organisms like Trypanosoma, guide RNAs direct the cutting of the mRNA. U bases are then added or removed, and the mRNA is joined back together.

Q31. What is translocation sequence? What is termination sequence?

Answer:

Translocation sequence: During translation, after the peptide bond is made, the ribosome shifts forward by one codon along the mRNA. This movement is called translocation. The factor **EF-G** (in bacteria) or **eEF-2** (in eukaryotes) uses GTP energy to push the ribosome forward. The term can also mean a signal sequence on a protein that tells the cell where to send it.

Termination sequence: This is a region on the DNA that tells the RNA polymerase to stop copying. In bacteria, there are two types: **(1)** A GC-rich hairpin followed by a string of U's (**rho-independent**), and **(2)** A signal that needs the **Rho protein** to pull the RNA off (**rho-dependent**). In eukaryotes, termination involves the poly-A signal and the **torpedo model**.

Q32. What are replication repressors?

Answer:

Replication repressors are proteins that control and limit how DNA or plasmids copy themselves.

In plasmids, they keep the number of copies in check. For example, in the ColE1 plasmid family, a small RNA called **RNA I** acts as a brake that stops the RNA primer needed to start replication.

In phages (like lambda phage), the **CI repressor protein** keeps the phage DNA quiet inside the bacteria (lysogenic state). It blocks the genes needed for the phage to multiply. If the repressor is destroyed (for example by UV damage), the phage starts copying and eventually bursts the cell.

Q33. What is transcriptional processing?

Answer:

Transcriptional processing (or transcript processing) means all the changes made to the raw RNA to turn it into a usable mRNA.

It has three main steps:

- 1. 5' Capping:** A 7-methylguanosine cap is put on the front end to protect it and help it get recognised.
- 2. Splicing:** Introns (useless bits) are cut out and exons (useful bits) are joined together by the spliceosome.
- 3. 3' Polyadenylation:** The back end is cut and a poly-A tail is added for protection and transport. These changes often happen while the RNA is still being made. The **CTD tail of RNA Pol II** helps bring in the right helper proteins at each step.

Q34. What are riboswitches? Functions of riboswitches.

Answer:

Riboswitches are special control regions found in the untranslated front part (5' UTR) of some bacterial mRNAs. They can directly bind small molecules without needing any protein.

Structure: A riboswitch has two parts: **(1) the aptamer** – which grabs the small molecule, and **(2) the expression platform** – which changes shape when the molecule binds.

Functions: When the small molecule binds, the riboswitch changes its 3D shape. This can either stop transcription (by forming a hairpin that acts as a stop signal) or stop translation (by hiding the ribosome binding site).

Example: A thiamine riboswitch controls genes that make thiamine. When there is enough thiamine, it binds the riboswitch and turns off the gene. This saves the cell from making too much.

Q35. Explain DNA processing events. RNA processing events.**Answer:**

DNA processing events: These include **DNA replication** (copying all the DNA), **DNA repair** (fixing mistakes or damage), **DNA recombination** (swapping pieces between DNA molecules), and **DNA methylation** (adding methyl groups to control genes).

RNA processing events (in eukaryotes):

1. **5' Capping:** A cap is added to the front end for protection.
 2. **Splicing:** Introns are removed and exons are joined together.
 3. **3' Polyadenylation:** A poly-A tail is added to the back end.
 4. **RNA editing:** Some bases in the mRNA may be changed (like C to U or A to I).
 5. **Transport:** The finished mRNA is sent from the nucleus to the cytoplasm.
- In prokaryotes, rRNA and tRNA are also trimmed and modified after being made.

Q36. What is replacement/displacement vector? Give info about insertion vectors. Two features of displacement vector.**Answer:**

Replacement vector: A type of cloning vector based on lambda phage. It has a middle piece of DNA (**stuffer fragment**) that is cut out and replaced with the DNA we want to clone. It can hold large inserts up to about **23 kb**.

Two features: (1) It has two restriction enzyme cutting sites around the stuffer fragment. (2) It can carry bigger DNA inserts than insertion vectors because space is freed up by removing the stuffer.

Insertion vector: This is also based on lambda phage but has only **one cutting site**. The foreign DNA is simply inserted into the phage DNA without removing anything. It carries smaller inserts (up to about **10 kb**) because the total size of the DNA cannot be too big for the phage to package.

Q37. What is the torpedo model of transcription termination?**Answer:**

The torpedo model explains how RNA Polymerase II stops making RNA in eukaryotes.

After the polymerase passes the **poly-A signal (AAUAAA)**, the mRNA is cut at the poly-A site. But the polymerase keeps going and makes extra RNA that has no cap on its front end.

A fast enzyme called **Xrn2** (in humans) or **Rat1** (in yeast) starts chewing this leftover RNA from the 5' end, like a torpedo chasing a ship.

When this enzyme catches up to the polymerase, it crashes into it and **knocks it off the DNA**. This ends transcription.

This model works together with the **allosteric model**, where the polymerase itself also becomes unstable after passing the poly-A signal.

Q38. Name any two proteins/enzymes for lactose metabolism. Their functions.**Answer:**

- 1. Beta-galactosidase** (made by the lacZ gene): This enzyme breaks lactose into **glucose and galactose** by cutting the bond between them. It also makes some **allolactose**, which is the molecule that switches on the lac operon.
- 2. Lactose permease** (made by the lacY gene): This is a **transport protein** in the cell membrane that pumps lactose from outside into the cell. Without it, lactose cannot get inside the bacterium.

Q39. Functions of Lac operon in lactose metabolism.

Answer:

The **lac operon** is a set of genes in E. coli that work together to use lactose as food when glucose is not available:

lacZ makes beta-galactosidase, which breaks lactose into glucose and galactose and also makes allolactose (the inducer).

lacY makes lactose permease, which brings lactose into the cell.

lacA makes transacetylase, which may help get rid of harmful sugar-like molecules.

Regulation: When there is no lactose, the repressor sits on the operator and blocks copying. When lactose is present, allolactose binds to the repressor and removes it, so the genes can be read. For maximum copying, glucose must also be absent – this raises cAMP levels, which activates the **CAP protein** to further boost transcription.

Q40. Explain how mRNA is moved from nucleus to the cytoplasm.

Answer:

After the mRNA is fully processed in the nucleus, it is sent to the cytoplasm through **nuclear pores**:

1. The 5' cap is grabbed by the **cap-binding complex (CBC)**, and the poly-A tail is held by poly-A binding proteins. These mark the mRNA as ready to go.
2. Export helper proteins, especially **NXF1/TAP**, attach to the mRNA. They know how to talk to the nuclear pore.
3. The **TREX complex** links the splicing process to export, loading onto the mRNA during splicing.
4. The mRNA (with all its attached proteins) passes through the **nuclear pore complex**.
5. On the other side, an enzyme called **Dbp5** uses energy to strip off the transport proteins, so the mRNA stays in the cytoplasm and is ready for the ribosome.

Q41. What is CTD? Role?

Answer:

CTD stands for **C-Terminal Domain**. It is a long tail at the end of the biggest part of RNA Polymerase II in eukaryotes.

It is made up of a repeating sequence of seven amino acids: **Tyr-Ser-Pro-Thr-Ser-Pro-Ser (YSPTSPS)**, repeated many times.

Roles:

1. It brings in the right helper proteins for RNA processing at the right time during transcription.

2. When **Serine-5** gets a phosphate group (added by TFIIF), capping enzymes are brought in to cap the mRNA.
3. When **Serine-2** gets a phosphate group (added by P-TEFb), splicing and poly-A tail factors are called in.
4. So the CTD works like a **landing pad** that links transcription to mRNA processing, making everything happen smoothly together.

Q42. Number of CTD heptapeptide repeats in yeast, C. elegans, Drosophila, and humans.

Answer:

The CTD repeat (YSPTSPS) is found different numbers of times in different organisms:

Yeast (*S. cerevisiae*): **26 repeats**

Worm (*C. elegans*): **32 repeats**

Fruit fly (*Drosophila*): **43 repeats**

Humans: **52 repeats**

More complex organisms generally have more repeats. Yeast needs at least **8 repeats** to survive. These repeats provide places for processing factors to attach.

Q43. What is meant by splice site finding/recognition errors?

Answer:

These are mistakes where the spliceosome picks the wrong place to cut and join the RNA:

1. **Exon skipping:** A real exon is accidentally left out, making the mRNA shorter and possibly useless.
2. **Cryptic splice site use:** A fake splice site (a sequence that looks like a real one but is not) is used instead. This adds wrong bits of intron or removes parts of an exon.
3. **Intron retention:** An intron is not cut out and stays in the mRNA, which usually makes the protein non-functional.

These errors can cause diseases like **beta-thalassemia** and **spinal muscular atrophy**. They are often caused by mutations in the important **GT-AG splice signals** or the branch point.

Q44. Termination of transcription by Rho protein / Role of Rho protein in transcription termination.

Answer:

Rho is a protein that helps stop transcription in bacteria. It works like a motor that chases the RNA polymerase.

How it works:

1. Rho attaches to a C-rich region on the mRNA called the **rut site** (Rho utilization site).
2. Using energy from ATP, Rho slides along the mRNA towards the polymerase (5' to 3' direction).
3. When the polymerase pauses or slows down, Rho catches up to it.

4. Rho then pulls apart the RNA from the DNA in the transcription bubble, **releasing the mRNA**.
 5. The polymerase falls off the DNA and transcription ends.
- About **20–50%** of genes in *E. coli* use this Rho-dependent way to end transcription.

Q45. What is pUC plasmid?

Answer:

pUC plasmids are a family of small cloning vectors used widely in labs.

Key features: (1) They are very small (about **2.7 kb**) and easy to work with. (2) They make many copies of themselves in the cell (about **500–700 copies**), which gives lots of DNA. (3) They have an **ampicillin resistance gene** for selecting bacteria that took up the plasmid. (4) They have a **multiple cloning site (MCS)** inside the lacZ gene, which allows **blue-white screening**.

How screening works: If foreign DNA is inserted into the MCS, it breaks the lacZ gene. Bacteria with the insert grow as **white colonies** on X-gal plates. Bacteria without the insert stay **blue**.

Q46. How is the D-loop in the secondary structure of tRNA named?

Answer:

The **D-loop** in tRNA is named after a special modified base called **dihydrouridine** (abbreviated as **D**) that is found in this loop.

Dihydrouridine is a changed form of uridine where the double bond between carbons 5 and 6 has been reduced.

In the **cloverleaf structure** of tRNA, the D-loop is one of four arms, along with the anticodon loop, the T-psi-C loop, and the acceptor stem. It helps keep the tRNA in its correct 3D shape and plays a part in how the tRNA is recognised by its matching enzyme.

Q47. Role of CTD phosphorylation in polymerase elongation.

Answer:

The **CTD tail of RNA Pol II** gets phosphate groups added and removed at different stages, and this controls elongation:

1. At the start, the CTD has **no phosphate groups**. This lets Pol II sit at the promoter.
2. **TFIIH** adds phosphate to **Serine-5**. This helps the polymerase leave the promoter and calls in **capping enzymes**.
3. As the polymerase starts elongating, **P-TEFb** adds phosphate to **Serine-2**. This is the signal for productive elongation.
4. The Ser-2 phosphorylated CTD brings in **splicing factors** and **poly-A tail factors**.
5. So the changing pattern of phosphate groups on the CTD acts like a **code** that tells the cell which stage of transcription is happening and which helpers are needed.

Q48. CRISPR stands for? CRISPR role/work + applications.

Answer:

CRISPR stands for **Clustered Regularly Interspaced Short Palindromic Repeats**.

How it works naturally: Bacteria store pieces of virus DNA between repeat sequences. When the same virus attacks again, the bacteria make a guide RNA from these stored pieces. This guide RNA leads the **Cas protein (like Cas9)** to cut the invading virus DNA.

As a lab tool (CRISPR-Cas9): Scientists design a guide RNA to match any gene they want. The Cas9 protein then cuts the DNA at that exact spot. The cell tries to fix the cut, and this can either disable the gene or let scientists insert new DNA.

Applications: Treating genetic diseases, making knockout animals for research, improving crops, diagnosing diseases, cancer research, and possible treatments for HIV.

Q49. What are spliceosomes? Functions of minor spliceosome.**Answer:**

Spliceosomes are big molecular machines made of RNA and proteins that cut out introns from pre-mRNA. They are built from five **snRNPs: U1, U2, U4, U5, and U6**, plus many other proteins.

The **major spliceosome** handles the common **GT-AG introns**, which make up over 99% of all introns.

Minor spliceosome: This is a different version that handles a rare type of intron called **AT-AC introns** (or U12-type introns). It uses different snRNPs: **U11, U12, U4atac, U5, and U6atac**.

Function of the minor spliceosome: It removes these rare introns found in about **700–800 human genes**. These genes are often involved in important jobs like DNA copying, RNA processing, and cell signalling.

Q50. What is transient excursion? Purpose of transient excursion.**Answer:**

Transient excursion means a brief, temporary opening or movement in a molecular structure during a biological process.

In transcription, it can refer to the short-lived opening of the DNA double helix at the promoter when RNA polymerase starts working. The DNA unwinds briefly to make the transcription bubble.

In translation, it can mean the brief shape changes in the ribosome as it shifts from one position to the next during translocation.

Purpose: These brief changes are needed for molecular machines to work properly. They allow short-term access to the DNA or RNA, help large complexes move along, and keep the processes of gene expression running smoothly.

Q51. Name two enzymes used in lactose metabolism.**Answer:**

1. Beta-galactosidase (lacZ product): Cuts lactose into glucose and galactose. It also makes **allolactose**, which switches on the lac operon.

2. Lactose permease (lacY product): A transporter in the cell membrane that brings lactose into the bacterial cell.

(A third enzyme, **transacetylase** from lacA, is also made but plays a smaller role.)

Q52. Names of RNA polymerases present in eukaryotes.

Answer:

Eukaryotes have three main RNA polymerases:

1. RNA Polymerase I (Pol I): Makes most ribosomal RNAs (28S, 18S, 5.8S rRNA). It works in the **nucleolus**.

2. RNA Polymerase II (Pol II): Makes messenger RNA (mRNA), most small nuclear RNAs (snRNA), and microRNAs. It is the most important one and has the **CTD tail**.

3. RNA Polymerase III (Pol III): Makes transfer RNA (tRNA), 5S rRNA, and some other small RNAs.

Plants also have two extra ones: **RNA Polymerase IV and V**, which help with gene silencing.

Q53. Briefly explain the Watson and Crick model of DNA.

Answer:

Watson and Crick described the structure of DNA in 1953. **The main points are:**

1. DNA is made of two strands twisted around each other in a **right-handed double helix**.

2. The two strands run in opposite directions (**antiparallel**) – one goes 5' to 3' and the other goes 3' to 5'.

3. The sugar-phosphate backbones are on the **outside**, and the bases point **inwards** like rungs of a ladder.

4. The bases pair up by **hydrogen bonds**: **A pairs with T** (2 bonds), and **G pairs with C** (3 bonds). This is called **complementary base pairing**.

5. The helix is **2 nm wide**, with **10 base pairs per full turn** (3.4 nm per turn). It has a major groove and a minor groove.

6. Because the bases are complementary, each strand can act as a template to copy the other during **replication**.

Q54. What is topoisomerase?

Answer:

Topoisomerases are enzymes that fix the twisting problems in DNA by cutting and re-joining the DNA strands.

Type I topoisomerase: Cuts **one strand** of DNA, lets it unwind to release the tension, and joins it back. It does not need ATP.

Type II topoisomerase (like **DNA gyrase** in bacteria): Cuts **both strands**, passes another part of the DNA through the gap, and seals it. It needs ATP. DNA gyrase can also add negative supercoils.

Why they are important: When DNA is being copied or read, the strands ahead get wound too tightly (supercoiled). Topoisomerases release this tension so that replication and transcription can keep going.

Q55. How do Group I introns splice DNA? What is spliced DNA? Role of introns in DNA.

Answer:

Group I intron splicing: Group I introns can remove themselves from the RNA without any protein help (**self-splicing**). Here is how:

1. A free **guanosine molecule** from outside attacks the 5' splice site, cutting the RNA at the start of the intron.
2. The free end of the first exon then attacks the 3' splice site, joining the two exons together. The intron is released.

Spliced DNA/RNA: This means the final nucleic acid from which introns have been removed and exons have been joined.

Role of introns: (1) They allow **alternative splicing**, so one gene can make many different proteins. (2) They contain **control elements** that help regulate genes. (3) They may help evolution by allowing **exon shuffling**. (4) Some introns carry useful small RNAs inside them.

Q56. What is CpG island?

Answer:

A **CpG island** is a stretch of DNA where there are many CG pairs next to each other (a C followed by a G, linked by a phosphate).

CpG islands are usually **300–3,000 base pairs** long and are found near the **promoters** of about 60–70% of human genes.

In most of the genome, the C in CpG is **methylated** (has a methyl group added). But in CpG islands near **active gene promoters**, the C is usually **not methylated**, which allows the gene to be read.

If a CpG island gets methylated when it should not be, the gene is **turned off**. Abnormal methylation is linked to **cancer** and other diseases.

Q57. What is a cosmid? Its role?

Answer:

A **cosmid** is a man-made cloning vector that is a mix of a **plasmid** and a **lambda phage**.

It has **cos sites** (from lambda phage) put into a plasmid. It also has an antibiotic resistance gene, an origin of replication, and restriction sites for inserting DNA.

Role: Cosmids are used to clone large DNA pieces (**37–45 kb**), which is much bigger than normal plasmids can handle. The cos sites let the DNA be packed into lambda phage heads in the lab, which then deliver the DNA into bacteria very well.

Once inside the cell, the cosmid copies itself like a normal plasmid. Cosmids are useful for making **genomic libraries** of large organisms.

Q58. What are classes of splicing? Their functions?**Answer:**

There are several types of splicing:

- 1. Major spliceosomal splicing:** Uses U1, U2, U4, U5, U6 snRNPs. Removes the common **GT-AG introns**. This is the standard splicing method for most genes.
- 2. Minor spliceosomal splicing:** Uses U11, U12, U4atac, U5, U6atac. Removes rare **AT-AC introns** found in a small number of genes.
- 3. Group I self-splicing:** The intron acts as its own enzyme. It uses an outside guanosine to cut and join the exons. Found in some rRNA and mitochondrial genes.
- 4. Group II self-splicing:** The intron forms a loop (lariat) using an internal A base. Found in mitochondria and chloroplasts. It is thought to be the ancestor of spliceosomal splicing.
- 5. tRNA splicing:** Special cutting and joining enzymes remove introns from pre-tRNA.

Q59. Why do peptide bonds not rotate freely?**Answer:**

Peptide bonds cannot rotate freely because they have a **partial double bond character**.

The bond between the carbon and nitrogen in a peptide bond shares electrons with the nearby oxygen. This sharing (called **resonance**) gives the C–N bond about **40% double bond nature**.

Because of this, the peptide bond is **flat and rigid** – all six atoms around it lie in the same plane.

Rotation is blocked around this bond. However, the bonds next to it (**phi and psi angles**) can rotate, which is how proteins can fold into their different shapes.

This rigidity is important because it limits how the protein chain can bend, helping to form regular structures like **alpha-helices and beta-sheets**.

Q60. What is photoreactivation experiment?**Answer:**

Photoreactivation is a light-dependent DNA repair method that directly fixes UV damage.

Mechanism: UV light makes neighbouring pyrimidine bases in DNA stick together (forming **pyrimidine dimers**). An enzyme called **photolyase** attaches to the dimer in the dark. When exposed to blue/visible light (300–500 nm), photolyase uses the light energy to **break the dimer apart** and fix the bases back to normal.

The experiment: Scientists damaged bacteria with UV light, which reduced their survival. They then put some bacteria in visible light and kept others in the dark. The bacteria in light survived much better because photolyase fixed their DNA. The bacteria kept in the dark could not repair the damage as well.

Photoreactivation is found in bacteria, fungi, and plants, but **not in humans**.

Q61. Differentiate between metabolomics and metagenomics.

Answer:

Metabolomics: The study of all small chemical molecules (metabolites) like sugars, fats, and amino acids in a cell or organism at one time. It uses tools like **mass spectrometry**. It helps in understanding diseases, finding drugs, and studying nutrition.

Metagenomics: The study of all the DNA collected directly from an environment (like soil, water, or the gut) without growing the organisms first. It uses **DNA sequencing**. It helps discover new microbes, enzymes, and understand microbial communities.

Key difference: Metabolomics looks at the **small chemical products** in a system. Metagenomics looks at the **total DNA** of all microbes living in an environment.

Q62. Genomics sequence of viruses.**Answer:**

Viral genomics means reading and studying the complete genetic material of viruses.

Viruses have many different types of genomes: they can be DNA or RNA, single-stranded or double-stranded, linear or circular, in one piece or in segments.

The Baltimore system groups viruses into 7 classes: **dsDNA** (Class I), **ssDNA** (Class II), **dsRNA** (Class III), **(+)ssRNA** (Class IV), **(-)ssRNA** (Class V), **ssRNA-RT** (Class VI, like HIV), and **dsDNA-RT** (Class VII, like Hepatitis B).

Why it matters: Sequencing virus genomes helps us find new viruses, track how they change (mutations), make vaccines, develop antiviral drugs, and follow disease outbreaks.

Q63. What is the role of sigma factor in protein synthesis initiation?**Answer:**

The sigma factor plays an indirect but key role in protein synthesis. It does not work on the ribosome directly but controls which mRNAs are made.

Role in transcription initiation: Sigma factor helps RNA polymerase find and bind to the right promoters on DNA, starting transcription. The mRNA produced then goes to the ribosome for translation into protein.

Different sigma factors turn on different sets of genes. For example, **sigma-70** makes mRNAs for everyday proteins. **Sigma-32** makes mRNAs for heat-shock proteins when the cell is stressed. So by choosing which genes to copy, sigma factors **decide which proteins** the cell will make.

Q64. How are second-generation biofuels produced? What are biofuels? Functions?**Answer:**

Biofuels: Fuels made from living things (plants, algae, etc.) instead of fossil fuels. They are renewable and produce less pollution.

First-generation biofuels come from food crops (like corn). **Second-generation** biofuels come from non-food plant waste (like straw, wood chips, grasses).

How second-generation biofuels are made:

1. Plant waste (like crop leftovers) is collected.

2. The tough plant material (cellulose, lignin) is broken down using heat or chemicals.
3. Enzymes (**cellulases**) turn the cellulose into simple sugars.
4. Microbes (like yeast) **ferment** the sugars into ethanol or butanol.
5. Other methods like gasification can also turn biomass into fuel.

Functions: They provide clean energy, reduce our need for oil, lower carbon emissions, and make good use of waste materials.

Q65. What is biodiesel?

Answer:

Biodiesel is a renewable fuel made from plant oils, animal fats, or used cooking oil.

How it is made: The oil or fat is mixed with an alcohol (usually methanol) and a catalyst (like sodium hydroxide) in a process called **transesterification**. This produces biodiesel (fatty acid methyl esters) and glycerol as a by-product.

It can be used in diesel engines on its own or mixed with regular diesel. Common mixes are **B20** (20% biodiesel) and **B100** (pure biodiesel).

Benefits: It is biodegradable, non-toxic, produces less pollution, and reduces our need for fossil fuels.

Q66. What is galactose operon (GAL)?

Answer:

The **GAL operon** is a group of genes that help bacteria (and yeast) break down galactose sugar.

In E. coli, it has three main genes:

galE – makes an enzyme that converts UDP-galactose to UDP-glucose.

galT – makes an enzyme that transfers a chemical group during galactose breakdown.

galK – makes galactokinase, which adds a phosphate to galactose.

These enzymes work together to turn galactose into a form that can enter the main energy pathway (**glycolysis**).

Control: The **GalR repressor** blocks the genes when galactose is absent. When galactose is present, it removes the repressor and the genes are turned on. If glucose is also present, the genes stay mostly off (**catabolite repression**).

In yeast, similar genes (GAL1, GAL7, GAL10) are controlled by the **Gal4 activator** and **Gal80 repressor**.

Q67. Types of interaction between microbe and plants.

Answer:

Microbes and plants interact in several ways:

1. **Mutualism (both win):** Mycorrhizal fungi help roots absorb nutrients like phosphorus, and the plant gives sugars in return. **Rhizobium** bacteria fix nitrogen for legumes.

2. **Commensalism (one benefits, other is unaffected):** Some bacteria live on plant surfaces and get food scraps without hurting the plant.

- 3. Parasitism (microbe wins, plant loses):** Some fungi, bacteria, and viruses cause plant diseases. For example, **Agrobacterium** causes tumours in plants.
- 4. Saprophytism:** Microbes feed on dead plant material and recycle nutrients back into the soil.
- 5. Endophytic interaction:** Some microbes live inside plant tissues without causing disease and may even help the plant fight infections or handle stress.

Q68. Types of Antimicrobials.

Answer:

Antimicrobials are substances that kill or stop the growth of germs. **The main types are:**

- 1. Antibiotics** (against bacteria): e.g., penicillin (blocks cell wall), tetracycline (blocks protein making), ciprofloxacin (blocks DNA copying).
 - 2. Antivirals** (against viruses): e.g., acyclovir (for herpes), oseltamivir (for flu), antiretrovirals (for HIV).
 - 3. Antifungals** (against fungi): e.g., amphotericin B (attacks fungal membrane), fluconazole (blocks membrane building).
 - 4. Antiparasitics** (against parasites): e.g., chloroquine (for malaria), metronidazole (for amoeba).
- They can be **bactericidal** (kill) or **bacteriostatic** (stop growth). They can be **broad-spectrum** (work on many types) or **narrow-spectrum** (target specific types).

Q69. What are SR proteins?

Answer:

SR proteins are a family of proteins that help with RNA splicing. They are named after their **serine-arginine (SR) rich region** called the RS domain.

Structure: They have one or two **RNA-binding parts (RRM)** at the front for grabbing RNA, and an **RS domain** at the back for interacting with other proteins.

Functions: (1) They bind to **exonic splicing enhancer (ESE)** sequences in exons to help the spliceosome find the right splice sites. (2) They bring in spliceosome parts like U1 and U2 snRNP. (3) They promote the inclusion of exons during alternative splicing. (4) They also help with mRNA export, translation, and mRNA stability.

Q70. Why are plasmids not included in the main cell's genome? / Why are plasmids extra-chromosomal?

Answer:

Plasmids stay separate from the main chromosome because:

- 1.** They copy themselves on their own using their own **origin of replication**.
- 2.** They do not carry **essential genes** – the cell can survive without them under normal conditions.
- 3.** They carry bonus genes (like antibiotic resistance) that only help in special situations.
- 4.** Many can **move between cells** (by conjugation), which would be hard if they were stuck in the chromosome.
- 5.** Being separate lets cells gain or lose them quickly to adapt to changing conditions.

Q71. What are plasmids? And where are they present?**Answer:**

Plasmids are small, circular pieces of double-stranded DNA that exist separately from the main chromosome.

Features: They copy themselves on their own. They carry 1–200 genes. They are not needed for normal life but carry useful extras like antibiotic resistance.

Where found: Mainly in **bacteria**. Also found in some archaea and yeast (e.g., the 2-micron plasmid in *S. cerevisiae*). Rarely in other organisms.

Types: **F-plasmid** (for mating/conjugation), **R-plasmid** (carry antibiotic resistance), **Col-plasmid** (make toxins called colicins), **Ti-plasmid** (cause plant tumours, found in *Agrobacterium*), and **degradative plasmids** (break down unusual chemicals).

Q72. What is meant by translocation?**Answer:**

Translocation has different meanings depending on the context:

In translation: It is the movement of the ribosome along the mRNA by one codon. After a peptide bond is made, the factor **EF-G** (bacteria) or **eEF-2** (eukaryotes) pushes the ribosome forward. The tRNAs shift: A-site to P-site, P-site to E-site, and a new codon appears at the A-site.

In genetics: It is a chromosome mutation where a piece of one chromosome breaks off and joins a different chromosome. Example: The **Philadelphia chromosome** (parts of chromosomes 9 and 22 swap) causes a type of blood cancer.

In cell biology: It can mean moving a protein across a membrane, like when proteins are pushed into the ER through a channel.

Q73. Translocation and its initiation subunits/elongation factors?**Answer:**

Translocation during translation uses special helper proteins called **elongation factors**:

In Bacteria:

EF-Tu: Brings the charged tRNA to the A-site using GTP energy.

EF-Ts: Recharges EF-Tu by swapping its used GDP for a fresh GTP.

EF-G: Does the actual translocation – pushes the ribosome forward one codon using GTP energy.

In Eukaryotes:

eEF-1A (same job as EF-Tu): Delivers charged tRNA to the A-site.

eEF-1B (same job as EF-Ts): Recharges eEF-1A.

eEF-2 (same job as EF-G): Drives translocation of the ribosome.

Note: Initiation factors (IF1, IF2, IF3 in bacteria; eIFs in eukaryotes) set up the ribosome at the start codon. They are separate from the elongation/translocation factors.

Q74. What is CPD / Cyclobutane Pyrimidine Dimers? CPD effects on DNA.

Answer:

CPDs are the most common type of DNA damage caused by UV light (especially UV-B).

How they form: When two pyrimidine bases next to each other (usually two thymines) absorb UV light, they become joined together by extra bonds, forming a four-sided ring (**cyclobutane ring**).

Effects on DNA:

1. The DNA helix gets **bent and distorted** at that point.
2. DNA polymerase **cannot read** past the joined bases, so replication is blocked.
3. RNA polymerase also gets stuck, **blocking transcription**.
4. If the damage is not fixed, it can cause **C to T mutations** when the cell tries to copy past it.
5. Too many CPDs in skin cells can lead to **skin cancer**.

Repair: By nucleotide excision repair (NER), photoreactivation (using photolyase and light), or base excision repair.

Q75. Note on CTD.**Answer:**

The CTD (C-Terminal Domain) is a long tail on the largest piece of RNA Polymerase II in eukaryotes.

Structure: It is made of repeating blocks of 7 amino acids: **Tyr-Ser-Pro-Thr-Ser-Pro-Ser**. The number of repeats varies: **26 in yeast, 32 in worms, 43 in fruit flies, and 52 in humans**.

How it works: The CTD switches between having phosphate groups and not having them. Without phosphates, Pol II can sit at the promoter. With **phosphate on Ser-5** (added by TFIIH), capping enzymes are called in. With **phosphate on Ser-2** (added by P-TEFb), splicing and poly-A factors arrive.

Importance: The CTD acts as a **platform** that coordinates transcription with all three mRNA processing events (capping, splicing, polyadenylation). It makes sure everything happens at the right time.

Q76. What do you know about the involvement of CTD tail in transcription and processing?**Answer:**

The CTD tail of RNA Pol II connects transcription to mRNA processing:

Initiation: The CTD must have no phosphates for Pol II to join the starting complex at the promoter.

Capping: TFIIH adds phosphate to **Ser-5** on the CTD. This brings in capping enzymes to put the 5' cap on the mRNA.

Elongation and splicing: **P-TEFb** adds phosphate to **Ser-2**. This change brings in splicing factors so introns can be removed while the RNA is still being made.

Polyadenylation: The Ser-2 phosphorylated CTD also brings in the enzymes that cut the mRNA and add the poly-A tail.

Termination: After the poly-A signal, the CTD helps bring in factors that stop transcription.

Recycling: After transcription ends, phosphatases (like FCP1) remove the phosphates from the CTD so Pol II can start a new round.

Q77. Functions of General Transcription Factors (GTFs).**Answer:**

GTFs are proteins that RNA Pol II needs to start transcribing any gene:

TFIID (contains TBP): Finds and sits on the TATA box in the promoter. TBP bends the DNA, marking where transcription should start.

TFIIA: Helps hold TFIID in place on the DNA and blocks repressors.

TFIIB: Acts as a bridge between TFIID and RNA Pol II. It helps position the polymerase at the right spot.

TFIIF: Comes in with RNA Pol II and helps it attach to the complex. Also helps melt the DNA.

TFIIE: Brings in TFIIH and helps with the switch from initiation to elongation.

TFIIH: Has two key activities: **(1) A helicase** that unwinds the DNA at the promoter to make the open complex, and **(2) A kinase** that adds phosphate to the CTD at Ser-5 to get transcription going.

Q78. Importance of gamma phage.**Answer:**

Gamma phage is a virus that infects **Bacillus anthracis** (the bacterium that causes anthrax).

Why it is important:

- 1. Diagnostic tool:** It is used in the **gamma phage lysis test** to confirm the identification of B. anthracis. If the bacteria are killed by gamma phage, it confirms they are B. anthracis.
- 2. Specificity:** It is very specific – it only attacks B. anthracis, not other similar Bacillus species. This makes it useful for identifying anthrax bacteria in labs and **biodefence**.
- 3. Research:** It helps scientists understand how phages and bacteria interact.

Q79. Importance of lambda phage.**Answer:**

Lambda phage is one of the most studied and useful viruses in molecular biology:

- 1. Model organism:** It is a key model for studying gene regulation, especially the switch between the **lytic** (killing) and **lysogenic** (quiet) life cycles.
- 2. Cloning vector:** Lambda phage DNA is used as a cloning vector. **Insertion vectors** can carry up to 10 kb, and **replacement vectors** can carry up to 23 kb of foreign DNA.
- 3. Gene regulation:** The **CI and Cro** regulatory system of lambda is a classic example of how proteins control genes.
- 4. Recombination:** It helped scientists understand site-specific recombination through its **integrase** enzyme.
- 5. Genomic libraries:** It is used to build genomic libraries because it can carry large inserts and deliver them into bacteria efficiently.

Q80. Importance of alpha phage as a vector.**Answer:**

Alpha phage (commonly referring to **M13 filamentous phage** in cloning) is used as a cloning vector because:

- 1. Single-stranded DNA production:** M13-based vectors produce **ssDNA**, which is useful for DNA sequencing (Sanger method), making probes, and site-directed mutagenesis.
- 2. Phagemid vectors:** Hybrid vectors combining M13 and plasmid features (like pBluescript) allow growth as either a plasmid (dsDNA) or phage (ssDNA).
- 3. Phage display:** M13 is widely used in **phage display technology**, where foreign proteins are shown on the phage surface. This is used to find new antibodies and drugs.
- 4. Non-lytic:** M13 does not kill its host cell. It keeps making phage particles that are released from the cell, making it easy to collect.

Q81. RS domain function in SR proteins.

Answer:

The **RS domain** (rich in arginine and serine) is at the back end of SR proteins. **Its jobs include:**

- 1. Protein-protein interactions:** It connects SR proteins to other splicing factors like U1-70K and U2AF, helping build the spliceosome.
- 2. Bridging splice sites:** When an SR protein sits on an exon, its RS domain reaches out to talk to U2AF at the nearby splice site. This helps the spliceosome recognise the exon.
- 3. Controlled by phosphorylation:** Kinases (like **SRPK1**) add phosphate groups to serines in the RS domain. This controls where the SR protein goes and how active it is.
- 4. Nuclear targeting:** The RS domain also acts as a signal that directs SR proteins into the nucleus, specifically to **nuclear speckles** where splicing factors are stored.

Q82. What changes occur in nucleosomes during transcription? Nucleosome remodelling.

Answer:

Nucleosomes pack DNA tightly, but they must be moved during transcription. **Changes include:**

- 1. Histone modification:** Chemical groups are added to histone tails. **Acetylation** (by HATs) loosens the DNA. **Methylation** can turn genes on or off depending on where it happens.
- 2. Chromatin remodelling:** ATP-powered complexes (like **SWI/SNF**) slide, eject, or rearrange nucleosomes to clear the promoter region.
- 3. Histone eviction:** During active transcription, histones are temporarily removed from in front of the moving polymerase and put back behind it by helper proteins (like the **FACT complex**).
- 4. Histone variant swap:** Normal histones are replaced with special ones (like **H2A.Z** or **H3.3**) that make nucleosomes less stable and easier to move.

All these changes are needed both to open the promoter and to let the polymerase travel through the gene.

Q83. Differentiate between prokaryotic and eukaryotic transcription machinery.

Answer:

Key differences:

RNA Polymerase: Bacteria have **only one**. Eukaryotes have **three main ones** (Pol I, II, III) for different RNA types.

Promoter: Bacteria use **-10 and -35 regions** read by sigma factor. Eukaryotes use the **TATA box** and other elements read by GTFs.

Transcription factors: Bacteria mainly need **sigma factor**. Eukaryotes need many GTFs (TFIIA, B, D, E, F, H) plus the **Mediator complex**.

Coupling: In bacteria, transcription and translation happen **at the same time**. In eukaryotes, they are **separated** by the nuclear membrane.

Processing: Bacterial mRNA is ready to use **straight away**. Eukaryotic mRNA must be **capped, spliced, and given a poly-A tail**.

Chromatin: Bacteria have **no histones**. Eukaryotes must **unpack chromatin** (nucleosomes) before genes can be read.

Q84. What are three events before translation?**Answer:**

Three things must happen before a protein can be made:

- 1. Transcription:** The gene is copied into mRNA by RNA polymerase. In eukaryotes, this happens in the nucleus.
- 2. mRNA processing (eukaryotes only):** The pre-mRNA gets a 5' cap, introns are spliced out, and a poly-A tail is added. Then the mature mRNA is sent to the cytoplasm.
- 3. Aminoacylation (tRNA charging):** Each tRNA must be loaded with its correct amino acid by the aminoacyl-tRNA synthetase enzyme using ATP energy. This ensures the right amino acids are available for the ribosome.

Q85. Lambda phage – comprehensive answer.**Answer:**

Lambda phage is a well-known virus that infects *E. coli*. It has linear double-stranded DNA (about **48.5 kb**) with sticky ends (**cos sites**).

Two life cycles: (1) **Lytic** – the phage copies its DNA, makes new phage particles, and bursts the cell open. (2) **Lysogenic** – the phage DNA integrates into the bacterial chromosome using **integrase** and sits quietly as a prophage.

Genetic switch: Two proteins control the choice: **CI repressor** (keeps phage quiet – lysogeny) and **Cro protein** (starts the killing cycle – lysis).

As a cloning vector: Lambda is used to clone large DNA pieces. Insertion vectors hold up to **10 kb**. Replacement vectors hold up to **23 kb**. The cos sites help pack the DNA into phage heads in the lab.

Applications: Making genomic libraries, studying gene regulation (the lysis/lysogeny switch is a textbook example), gene mapping, and developing other vectors like cosmids.

Q86. UV radiation types and their effects on life.

Answer:

UV radiation comes in three types:

UV-A (315–400 nm): Least harmful but goes deep into skin. Causes ageing, wrinkles, and some skin cancers by making harmful **free radicals (ROS)**.

UV-B (280–315 nm): More harmful. Causes sunburn and directly damages DNA by fusing pyrimidine bases together (making **CPDs**). Main cause of **skin cancer**. Most UV-B is blocked by the ozone layer.

UV-C (100–280 nm): Most dangerous but fully blocked by the ozone layer. Used in labs and hospitals to **kill germs** on surfaces and in water.

Overall effects: DNA damage, mutations, skin cancer, eye damage (cataracts), weakened immune system, harm to sea life and plankton, and reduced plant growth.

Q87. Differentiate between charged and uncharged amino acids (tRNA).**Answer:**

This question is about charged and uncharged tRNA:

Charged tRNA (aminoacyl-tRNA): A tRNA that has its amino acid attached to the 3' CCA end. It was loaded by the aminoacyl-tRNA synthetase enzyme using ATP. It is ready to go to the ribosome and deliver its amino acid for protein building.

Uncharged tRNA (deacylated tRNA): A tRNA with no amino acid attached. This happens after the tRNA has given its amino acid to the growing protein chain. It leaves the ribosome from the E-site and needs to be recharged before it can work again.

Key difference: Charged = has amino acid, ready for translation. Uncharged = empty, needs reloading.

The balance between charged and uncharged tRNA in the cell can trigger stress responses, like the **stringent response** in bacteria when amino acids are running low.

Q88. What is catabolite repression?**Answer:**

Catabolite repression (also called the **glucose effect**) is a system where the presence of glucose stops the cell from using other sugars like lactose.

How it works: When glucose is available, the level of a signal molecule called **cAMP** stays low. The **CAP protein** (Catabolite Activator Protein) needs cAMP to become active. Without cAMP, CAP cannot bind to DNA, and genes for other sugars are barely turned on.

When glucose runs out, **cAMP levels go up**. cAMP binds to CAP, and the CAP-cAMP pair binds to DNA near the promoter, helping RNA polymerase bind and greatly boosting transcription.

So for the lac operon to be fully on, two conditions must be met: **lactose present** (to remove the repressor) AND **glucose absent** (to activate CAP-cAMP).

Q89. What is catabolite repression?**Answer:**

(Same as Q88)

Catabolite repression is a control system where glucose stops the expression of genes for using other sugars. It works through **cAMP and CAP**: low glucose = high cAMP = CAP active = genes on. High glucose = low cAMP = CAP inactive = genes stay off.

Q90. Write the regulations of galactose in an operon.

Answer:

The **GAL operon** is regulated by both negative and positive control:

Negative control: The **GalR repressor** binds to two operator sites (OE and OI) on either side of the promoter. It creates a DNA loop that blocks RNA polymerase from reaching the genes.

Induction: When galactose enters the cell, its modified form binds to GalR and changes its shape, making it let go of the DNA. The loop opens and the genes can be copied.

Catabolite repression: Just like the lac operon, the GAL operon also needs low glucose for full activity. Low glucose means high cAMP, which activates **CAP** to help RNA polymerase bind.

Full activation needs: galactose present (repressor removed) + glucose absent (CAP-cAMP active).

Q91. Write two 3' to 5' RNA degradation mechanisms.

Answer:

1. Exosome pathway: The **exosome** is a big protein complex that chews up RNA from the 3' end towards the 5' end. In eukaryotes, after the poly-A tail is removed, the exosome eats the mRNA from the back. Any leftover cap is then removed by the **DcpS enzyme**.

2. PNPase pathway: In bacteria, an enzyme called **polynucleotide phosphorylase (PNPase)** breaks down RNA from the 3' end. It is part of a bigger machine called the **degradosome**, which also includes the endonuclease RNase E and a helicase called RhlB.

Both systems help control mRNA levels, remove faulty RNA, and recycle the building blocks.

Q92. Steps of attaching amino acids to tRNA.

Answer:

Aminoacylation (tRNA charging) is done by aminoacyl-tRNA synthetase in **two steps**:

Step 1 – Activation: The amino acid reacts with ATP to form **aminoacyl-AMP** (an activated form). Pyrophosphate (PPi) is released and quickly broken down, which makes this step one-way.

Step 2 – Transfer: The activated amino acid is moved from aminoacyl-AMP to the **3' end** of the matching tRNA (the CCA tail). AMP is released.

Overall: Amino acid + tRNA + ATP → Aminoacyl-tRNA + AMP + 2Pi

Each of the 20 synthetases also has a **proofreading ability** to check that the right amino acid is on the right tRNA. This accuracy is vital for making correct proteins.

Q93. What are SR proteins? Role of SR proteins in splicing.

Answer:

SR proteins are important splicing helper proteins. They have RNA-binding parts (**RRM**) and a serine-arginine-rich domain (**RS domain**).

Roles in splicing:

1. They find splice sites by sitting on **exonic splicing enhancer (ESE)** sequences within exons.
2. They call in spliceosome parts: their RS domain brings **U1 snRNP** to the 5' splice site and **U2AF** to the 3' splice site.
3. They help define exons by connecting the factors on either side of each exon.
4. They control **alternative splicing** – different SR proteins push for including or skipping certain exons. The balance between SR proteins and hnRNP proteins decides the splicing pattern.
5. They also help with mRNA export, nonsense-mediated decay, and translation.

Q94. Role of 8-oxoguanine.**Answer:**

8-Oxoguanine (8-oxoG) is one of the most common DNA damages caused by reactive oxygen species (**ROS**).

How it forms: ROS attack the base guanine and add an oxygen atom, creating 8-oxoguanine.

The problem: 8-oxoG can mispair with **adenine** (instead of its normal partner cytosine). If this is not fixed before replication, it causes a **G:C to T:A mutation**.

How the cell fixes it (GO system in E. coli):

MutM: Removes 8-oxoG when it is paired with C (before damage spreads).

MutY: Removes the wrong A when 8-oxoG has already mispaired with A.

MutT: Destroys 8-oxo-dGTP in the free nucleotide pool so it cannot be used during copying.

In humans, similar enzymes (**OGG1, MUTYH, MTH1**) do the same jobs. Faults in these repair systems are linked to **cancer**.

Q95. Ribosome sites A, E, P.**Answer:**

(Same as Q7 and Q8 – brief summary)

A-site (Aminoacyl): The new tRNA with its amino acid enters here. Codon-anticodon matching happens here.

P-site (Peptidyl): Holds the tRNA with the growing protein chain. Peptide bonds are made here.

E-site (Exit): Holds the empty tRNA as it leaves the ribosome.

tRNAs move through in order: **A → P → E**, driven by EF-G/eEF-2 and GTP energy.

Q96. What is open complex formation in transcription?**Answer:**

Open complex formation is when the double-stranded DNA at the promoter unwinds to expose the template strand so RNA can be made.

In bacteria: After the RNA polymerase holoenzyme (with sigma factor) binds to the promoter (forming the closed complex), the sigma factor helps melt open about **12–14 base pairs** around the -10 region, creating the open complex.

In eukaryotes: The transcription factor **TFIIH** uses its helicase activity and ATP energy to unwind the DNA at the promoter after the pre-initiation complex is assembled.

Why it matters: RNA polymerase can only use **single-stranded DNA** as a template. Without the open complex, transcription cannot begin.

Q97. Describe elongation in transcription.

Answer:

(Similar to Q21 – with extra points)

Elongation is when RNA polymerase moves along the DNA and builds the RNA chain:

1. It reads the template strand **3' to 5'** and makes RNA **5' to 3'**.
2. A bubble of about 12–14 bp moves with the enzyme. About 8–9 bases of RNA stay paired with the DNA inside the bubble.
3. NTPs (ATP, GTP, CTP, UTP) are matched to the template and added one by one.
4. **Topoisomerases** fix the twisting that builds up ahead of and behind the polymerase.
5. In eukaryotes, helper factors like **TFIIS** help the polymerase get past difficult spots. The **FACT complex** helps move nucleosomes out of the way.
6. In eukaryotes, capping and splicing happen during elongation.
7. Speed: about **40–80 bases/second** in bacteria, **20–40** in eukaryotes.

Q98. Write two errors in splice site recognition. What are splice site finding errors?

Answer:

Splice site finding errors are mistakes where the spliceosome cuts or joins the RNA in the wrong place:

Error 1 – Exon skipping: A real exon is missed and removed along with the introns. This makes the mRNA shorter and may produce a broken protein. Example: some cases of **Duchenne muscular dystrophy**.

Error 2 – Cryptic splice site use: A sequence that looks like a real splice site (but is not) gets used instead. This can add extra intron sequence or cut out part of an exon. Example: mutations in the beta-globin gene cause this, leading to **beta-thalassemia**.

Other errors include **intron retention** (intron stays in the mRNA) and wrong 5' or 3' splice site choice. These errors often come from mutations in the conserved **GT/AG signals** or the branch point.

Q99. Role of RNA Polymerase II in splicing. How is RNA Pol II different from other types?

Answer:

Role in splicing: RNA Pol II is special because its **CTD tail** directly calls in splicing factors while the mRNA is still being made. This is called **co-transcriptional splicing**. The phosphate pattern on the CTD (Ser-2 and Ser-5) decides which factors are brought in at each stage.

How Pol II differs from Pol I and Pol III:

Pol I: Makes large ribosomal RNAs in the nucleolus. Has no CTD.

Pol II: Makes mRNA and most snRNAs. Has the **CTD**. Needs many GTFs. Its products get capped, spliced, and given a poly-A tail.

Pol III: Makes tRNA, 5S rRNA, and other small RNAs. Has no CTD like Pol II.

Key point: Only Pol II has the CTD, and only its products go through all three processing steps (capping, splicing, polyadenylation).

Q100. What is allostery?

Answer:

Allostery is when a protein's activity is changed by a molecule binding to a different spot (not the active site). This other spot is called the **allosteric site**.

When something binds to the allosteric site, it changes the shape of the protein, making it more active or less active.

Positive allosteric effector: Makes the protein work better.

Negative allosteric effector: Slows the protein down or stops it.

Examples: (1) Allolactose binds to the lac repressor at its allosteric site, changing its shape so it falls off the DNA. (2) Haemoglobin shows allostery – when one subunit grabs oxygen, the others become better at grabbing oxygen too. (3) ATCase enzyme is slowed down by CTP (end product) and sped up by ATP.

Q101. Differentiate between nucleoside and nucleotide.

Answer:

Nucleoside = a base + a sugar (no phosphate). **Examples:** adenosine, guanosine, cytidine.

Nucleotide = a base + a sugar + one or more phosphate groups. **Examples:** AMP, dGTP.

Nucleotides are the building blocks of DNA and RNA.

Key difference: A nucleotide is just a nucleoside with phosphate groups attached to the 5' carbon of the sugar.

Nucleotides join together by **phosphodiester bonds** to make the long chains of DNA and RNA.

Q102. How is RNA Polymerase II different from other types?

Answer:

(Same as part of Q99 – summary)

Pol II stands out because: (1) It copies protein-coding genes into mRNA. (2) It has a unique **CTD tail** that coordinates mRNA processing. (3) It needs the most helper proteins (TFIIA, B, D, E, F, H + Mediator). (4) Its products are the only ones that get all three processing steps: **capping, splicing,**

and polyadenylation. (5) It is poisoned by **alpha-amanitin** (mushroom toxin) at low amounts, while Pol I is resistant and Pol III needs a higher dose.

Q103. Name all four primary components of the machinery responsible for translating mRNA to proteins.

Answer:

The four main parts of the translation machinery are:

- 1. mRNA:** Carries the genetic instructions from DNA. It has codons (groups of 3 bases) that specify which amino acids to add.
- 2. Ribosomes:** The machines where proteins are assembled. They have a large and small part (70S in bacteria, 80S in eukaryotes) with A, P, and E sites.
- 3. tRNA:** Adapter molecules that carry amino acids to the ribosome. Each tRNA has an anticodon that matches an mRNA codon.
- 4. Aminoacyl-tRNA synthetases:** Enzymes that load each tRNA with the correct amino acid. There are 20 of them (one per amino acid), and they make sure translation is accurate.

Q104. Explain translocation. At which subunit does it occur?

Answer:

Translocation is the step where the ribosome shifts forward by one codon along the mRNA.

How it happens: After the peptide bond is formed, **EF-G** (bacteria) or **eEF-2** (eukaryotes) binds with GTP. The energy from GTP breakdown pushes the ribosome forward one codon.

Result: The peptidyl-tRNA moves from **A-site to P-site**. The empty tRNA moves from **P-site to E-site**. A new empty A-site is ready for the next tRNA.

Which subunit: Translocation mainly involves the **large subunit** (50S in bacteria, 60S in eukaryotes). The peptide bond-forming centre and the GTPase centre (where EF-G works) are both on the large subunit. However, both subunits move together in a **ratchet-like motion**.

Q105. Describe and explain different ways through which alternative splicing can generate variety of mRNA sequences.

Answer:

Alternative splicing makes different mRNAs from one gene through these patterns:

- 1. Exon skipping:** An exon is either included or left out. This is the most common type. Different tissues may include or skip a specific exon.
- 2. Alternative 5' splice site:** The same intron has two or more possible starting points for the cut, making the upstream exon longer or shorter.
- 3. Alternative 3' splice site:** Two or more ending points exist for cutting the intron, changing the length of the downstream exon.
- 4. Intron retention:** An intron is not removed and stays in the final mRNA. This may add new coding information or create an early stop codon.

5. Mutually exclusive exons: Two or more neighbouring exons are spliced so only one is kept at a time, never both.

These patterns let one gene make many different protein versions. The choice is controlled by **SR proteins**, **hnRNPs**, and tissue-specific splicing factors.

Q106. mRNA has a short life span. How can a particular mRNA and its life span be analyzed? Name and description of the experiment.

Answer:

Experiment name: Pulse-Chase Analysis

Description:

- 1. Pulse:** Cells are given radioactive uridine (like ^3H -uridine) for a short time. This labels all the RNA being made during that period.
 - 2. Chase:** The radioactive material is removed and replaced with normal (unlabelled) uridine. From now on, only the already-labelled RNA is tracked.
 - 3. Sampling:** At regular intervals, RNA is taken out of the cells.
 - 4. Detection:** The amount of the specific labelled mRNA is measured using **Northern blot, autoradiography, or RT-qPCR.**
 - 5. Result:** The time it takes for the mRNA level to drop to half is called its **half-life.**
- Another method:** Block all new RNA production using **actinomycin D** (a transcription blocker), then measure how fast the existing mRNA disappears over time.

✧ **End of Document** ✧

May Allah bless your efforts and make your path easy.

When we lift others, Allah lifts us. 🤝